

Solutions to JEE Advanced Booster Test – 4 | 2022 | Code A

[PHYSICS]

1.(BC) Let the length of the cylinder below the surface in equilibrium be x_0

Then, Weight = Buoyant force

$$\Rightarrow \rho(AL)g = (3\rho)(Ax_0)g \Rightarrow x_0 = \frac{L}{3}$$

Since the cylinder is initially at rest, the initial position must be its upper extreme position. And since the equilibrium is a distance x_0 below the initial position, the amplitude of the SHM is x_0 itself

Now, if the cylinder is displaced by a distance x upward from the equilibrium position, the buoyant force on it decreases by a magnitude $\Delta F_B = (3\rho)(Ax)g = 3\rho Agx$

Since the cylinder was in equilibrium, and the upward acting buoyant force has decreased in magnitude, the net force on the cylinder is now ΔF_B downwards

$$\text{Therefore, the time period of the SHM is } T = 2\pi\sqrt{\frac{\rho AL}{3\rho Ag}} = 2\pi\sqrt{\frac{L}{3g}}$$

2.(BC) Rate of heat loss due to radiation, $\frac{dQ}{dt} = \sigma A(T^4 - T_0^4)$

$$\text{Therefore, } \left(\frac{dQ}{dt}\right)_1 = \frac{\sigma(4\pi R^2)((3T_0)^4 - T_0^4)}{\sigma\left(4\pi\left(\frac{3R}{2}\right)^2\right)((2T_0)^4 - T_0^4)} = \frac{64}{27}$$

$$\text{Rate of cooling, } \frac{dT}{dt} = \frac{\left(\frac{dQ}{dt}\right)}{mC}$$

Here, m is mass and C is specific heat

$$\text{Therefore, } \left(\frac{dT}{dt}\right)_1 = \left(\frac{\left(\frac{dQ}{dt}\right)_1}{\left(\frac{dQ}{dt}\right)_2}\right)\left(\frac{m_2}{m_1}\right) = \left(\frac{64}{27}\right)\left(\frac{\left(\frac{3R}{2}\right)^3}{R^3}\right) = 8$$

3.(AC) After it is released, the block A oscillates with angular frequency $\omega = \sqrt{\frac{k}{m}}$ and amplitude $2d$

$$\text{Therefore, its velocity just before collision is } v_0 = \omega\sqrt{(2d)^2 - d^2} = \sqrt{\frac{3k}{m}}d$$

After the collision, let the velocities of A and B be v_1 and v_2 , both towards right

Conserving momentum, $mv_0 = mv_1 + 2mv_2$

From Newton's experimental law, $v_2 - v_1 = v_0$

Solving, we get

$$v_1 = -\frac{1}{3}v_0 \quad \text{and} \quad v_2 = \frac{2}{3}v_0$$

$$\text{Therefore, } v_1 = -\sqrt{\frac{k}{3m}}d \quad \text{and} \quad v_2 = \sqrt{\frac{4k}{3m}}d$$

Now, let the amplitude of A after the collision be A_0

$$\text{Then, } v_1^2 = \frac{k}{m}(A_0^2 - d^2) \Rightarrow \frac{kd^2}{3m} = \frac{k}{m}(A_0^2 - d^2) \Rightarrow A_0 = \frac{2d}{\sqrt{3}}$$

4.(BCD) Work done during an adiabatic process,

$$W = -nC_V \Delta T \Rightarrow W = -(1) \left(\frac{3}{2} R \right) (T_f - T_0) = \frac{3RT_0}{4} \Rightarrow T_f = \frac{T_0}{2}$$

For an adiabatic process, $P^{1-\gamma} T^\gamma = \text{constant}$

$$\Rightarrow P_f = P_0 \left(\frac{T_0}{T_f} \right)^{\frac{\gamma}{1-\gamma}} = P_0 (2)^{-5/2} = \frac{P_0}{4\sqrt{2}}$$

Average kinetic energy of a gas molecule, $K_{av} = \frac{3}{2} KT$

We can see that K_{av} has halved.

For an isobaric process, $\Delta Q = nC_p \Delta T = (1) \left(\frac{5R}{2} \right) \left(T_0 - \frac{T_0}{2} \right) = \frac{5RT_0}{4}$

5.(ABD) Total internal energy, $U_{Total} = U_1 + U_2 = \frac{3}{2}(1)(R)(460) + \frac{5}{2}(1)(R)(300) = 1440R$

Conserving internal energy, $U_{mix} = U_1 + U_2$

$$\Rightarrow \frac{3}{2}(1)RT_{mix} + \frac{5}{2}(1)RT_{mix} = \frac{3}{2}(1)(R)(460) + \frac{5}{2}(1)(R)(300) \Rightarrow T_{mix} = 360 \text{ K}$$

Now, root mean square speed, $v_{rms} = \sqrt{\frac{3RT}{M}}$

$$\text{Therefore, } \frac{(v_{rms})_{Argon}}{(v_{rms})_{Nitrogen}} = \sqrt{\frac{M_{Nitrogen}}{M_{Argon}}} = \sqrt{\frac{28}{40}} = \sqrt{\frac{7}{10}}$$

Also, total kinetic energy, $K_{Total} = \frac{f}{2} nRT$ (f is the degree of freedom)

$$\text{Therefore, } \frac{(K_{Total})_{Argon}}{(K_{Total})_{Nitrogen}} = \frac{3}{5}$$

6.(AD) Volume of the liquid that spills out

$$\Delta V = V_0(1 + 4\gamma\Delta T) - V_0(1 + \gamma\Delta T) = 3V_0\gamma\Delta T$$

Now, let the area of cross-section of the beaker at temperature T_0 be A_0

Then, we know that $V_0 = A_0 H_0$

Therefore, the area of cross-section of the beaker at temperature $T_0 - \Delta T$ is $A = A_0 \left(1 - \left(\frac{2\gamma}{3} \right) \Delta T \right)$

[Since the area thermal expansion coefficient of any material is $\frac{2}{3}$ times its volume thermal expansion coefficient]

Now, the volume of the liquid remaining in the beaker at temperature $T_0 + \Delta T$ is equal to the capacity of the beaker at this temperature, i.e. $V_0(1 + \gamma\Delta T)$.

Therefore, the volume of this liquid in the beaker at temperature $T_0 - \Delta T$ becomes

$$V = V_0(1 + \gamma\Delta T)(1 - (4\gamma)(2\Delta T)) = V_0(1 - 7\gamma\Delta T)$$

So, the height of the liquid column in the beaker at temperature $T_0 - \Delta T$ is

$$H = \frac{V}{A} = \frac{V_0(1 - 7\gamma\Delta T)}{A_0 \left(1 - \frac{2}{3}\gamma\Delta T \right)} = \left(\frac{V_0}{A_0} \right) \left(1 - \frac{19}{3}\gamma\Delta T \right) = H_0 \left(1 - \frac{19}{3}\gamma\Delta T \right)$$

$$\text{Therefore, } \Delta H = \frac{19}{3} H_0 \gamma \Delta T$$

- 7.(D) The temperature of the gas rises during the isochoric heating (process I) and the isobaric expansion (process II), and it decreases during the isochoric cooling (process III) and the isobaric compression (process IV). Therefore, the gas reaches its maximum temperature when its pressure and volume are αP_0 and βV_0 respectively.

From the ideal gas equation, the maximum temperature, $T_{\max} = \frac{(\alpha P_0)(\beta V_0)}{P_0 V_0} T_0 = \alpha \beta T_0$

- 8.(C) In the isochoric cooling, the change in internal energy,

$$\Delta U = n C_V \Delta T = \left(\frac{C_V}{R} \right) (\beta V_0) (P_0 - \alpha P_0) = -\frac{3}{2} (\alpha - 1) \beta P_0 V_0$$

- 9.(C) Let $(\Delta \ell)_P$ be the change in length due to thermal expansion in rod 'P'

$$(\Delta \ell)_P = \ell \alpha \Delta T \quad \dots (i)$$

Let $(\Delta L)_P$ is change in length due to elastic force in rod 'P' $\Rightarrow (\Delta L)_P = \frac{F \ell}{AY} \quad \dots (ii)$

Similarly in rod 'Q', we can consider $(\Delta \ell)_Q$ and $(\Delta L)_Q$ change in length due to thermal expansion and elastic forces.

As net change in total length of rod $(P + Q) = 0$

Hence $(\Delta \ell)_P + (\Delta L)_P + (\Delta \ell)_Q + (\Delta L)_Q = 0$

$$\ell \alpha \Delta T + \frac{F \ell}{AY} + (2\ell)(2\alpha \Delta T) + \frac{F(2\ell)}{A(2Y)} = 0$$

$$\left(\frac{2F \ell}{AY} \right) = -5\ell \alpha \Delta T \quad ; \quad F = -\frac{5AY \alpha \Delta T}{2}$$

- 10.(B) Net change in length of rod P or rod Q is going to be the shift in junction

$$\Rightarrow \text{Shift} = (\Delta \ell)_P + (\Delta L)_P$$

$$= \ell \alpha \Delta T + \frac{F \ell}{AY} = \ell \alpha \Delta T - \frac{5AY \alpha \Delta T \ell}{2AY} = \ell \alpha \Delta T - \frac{5}{2} \ell \alpha \Delta T = -\frac{3}{2} \ell \alpha \Delta T$$

(-ve sign mean rod 'P' is overall compressed)

- 11.(B) Time period of the given SHM, $T = \frac{2\pi}{\omega}$

In general, to find the time interval between two events during an SHM, first find the phase angles associated with those two events, say ϕ_1 and ϕ_2 . Then, the time interval between the events will be:

$$\Delta t = \frac{T}{2\pi} (\phi_2 - \phi_1)$$

So, for the SHM $x(t) = A \sin \left(\omega t + \frac{\pi}{3} \right)$, the particle is at phase angle $\phi_1 = \frac{\pi}{3}$ at $t = 0$ i.e. at $x = \frac{\sqrt{3}}{2} A$ and going towards the right extreme ($x = +A$)

- (i) The particle will be at its mean for the first time when it is at phase angle $\phi_2 = \pi$

$$\text{Therefore,} \quad \Delta t = \frac{T}{2\pi} (\phi_2 - \phi_1) = \frac{T}{2\pi} \left(\pi - \frac{\pi}{3} \right) = \frac{T}{3} = \frac{2\pi}{3\omega}$$

- (ii) The particle will have the same speed as at $t = 0$ when it is at the same position as at $t = 0$, i.e.

$$\text{at phase angle } \phi_2 = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$$

$$\text{Therefore,} \quad \Delta t = \frac{T}{2\pi} (\phi_2 - \phi_1) = \frac{T}{2\pi} \left(\frac{2\pi}{3} - \frac{\pi}{3} \right) = \frac{T}{6} = \frac{\pi}{3\omega}$$

- (iii) The particle is at rest when it is at its right extreme, i.e. at phase angle $\phi_2 = \frac{\pi}{2}$

$$\text{Therefore,} \quad \Delta t = \frac{T}{2\pi} (\phi_2 - \phi_1) = \frac{T}{2\pi} \left(\frac{\pi}{2} - \frac{\pi}{3} \right) = \frac{T}{12} = \frac{\pi}{6\omega}$$

$$(iv) \quad a = \frac{a_{\max}}{2} \Rightarrow \omega^2 x = \pm \frac{\omega^2 A}{2} \Rightarrow x = \pm \frac{A}{2}$$

So, the first time this will happen after $t = 0$ is when the particle is at $x = \frac{A}{2}$.

Since at $t = 0$ the particle is at $x = \frac{\sqrt{3}}{2}A$, the first time that it will be at $x = \frac{A}{2}$ is at phase angle

$$\phi_2 = \frac{5\pi}{6}$$

$$\text{Therefore,} \quad \Delta t = \frac{T}{2\pi}(\phi_2 - \phi_1) = \frac{T}{2\pi}\left(\frac{5\pi}{6} - \frac{\pi}{3}\right) = \frac{T}{4} = \frac{\pi}{2\omega}$$

12.(D) Use $\Delta Q = \Delta U + W$

13.(4) Let the mass of the block and the bullet be M and m respectively

Let the velocity of the bullet just before it strikes the block be v_0

Then, by momentum conservation, the velocity of the block-bullet system just after the bullet gets embedded is

$$v = \left(\frac{m}{M+m}\right)v_0$$

Now, the initial position of the block was an equilibrium position. Since the mass of the bullet is negligible compared to the block, this initial position can be considered equilibrium for the block-bullet system as well. Therefore, this initial position is the mean position of the subsequent SHM, and so the velocity of the system after the embedding must be the maximum velocity during SHM, i.e. $A\omega$.

Now, the angular frequency of the system,

$$\omega = \sqrt{\frac{k}{M+m}}$$

$$\text{Since } v = A\omega, \quad \left(\frac{m}{M+m}\right)v_0 = A\left(\sqrt{\frac{k}{M+m}}\right)$$

Therefore, the amplitude of the SHM,

$$A = \frac{mv_0}{\sqrt{(M+m)k}} \approx \frac{mv_0}{\sqrt{Mk}} = \frac{(0.048)(50)}{\sqrt{(12)(300)}} = 0.04 \text{ m} = 4 \text{ cm}$$

Now, since the initial position is the mean position of the SHM, and the point where the system first comes to rest is the upper extreme position, the height difference between them is simply the amplitude of the SHM, i.e. 4 cm.

14.(800) Let the mass of ice needed be x

We know that

Heat absorbed by ice to rise to 0°C

=

(Heat lost by calorimeter to fall to 0°C + Heat lost by water to fall to 0°C and freeze)

$$\text{Therefore,} \quad x(0.5)(0 - (-40)) = 50(1)(20 - 0) + 150(1)(20 - 0) + 150(80)$$

$$\text{Solving, we get} \quad x = 800 \text{ g}$$

15.(0.33) We are given that $\frac{C_p}{C_v} = \frac{3}{2}$

We already know that $C_p - C_v = R$

Solving, we get $C_p = 3R$ and $C_v = 2R$

In an isobaric process, $W = P\Delta V$

$$\Delta Q = nC_p\Delta T = \left(\frac{C_p}{R}\right)P\Delta V \quad \text{Therefore,} \quad \frac{W}{\Delta Q} = \frac{R}{C_p} = \frac{1}{3}$$

16.(20) According to Newton's law of cooling, the time taken by a body to cool from temperature T_1 to temperature T_2 in a room at temperature T_0 is proportional to $\log_e \left(\frac{T_1 - T_0}{T_2 - T_0} \right)$

Therefore, using the given information, $\log_e \left(\frac{60 - T_0}{50 - T_0} \right) = \log_e \left(\frac{40 - T_0}{35 - T_0} \right)$

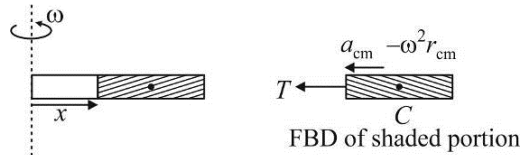
Solving, we get $T_0 = 20^\circ\text{C}$

17.(0.33) Distance of the CM of the shaded portion from the axis $= x + \left(\frac{\ell - x}{2} \right) = \frac{\ell + x}{2}$

Therefore,

$$T = m'\omega^2 r_{cm} = \frac{m}{\ell}(\ell - x)\omega^2 \left(\frac{\ell + x}{2} \right)$$

$$\Rightarrow T(x) = \frac{m\omega^2}{2\ell}(\ell^2 - x^2)$$

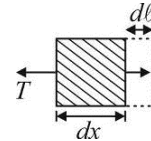


Since, tension depends on x , different parts of rod are under different stress and so strain will be different. Consider an element of length dx located at a distance x from the axis

$$\text{Stress in the element} = \frac{T}{A}$$

$$\Rightarrow \text{Strain} = \frac{\text{Stress}}{Y} = \frac{T}{AY}$$

$$\Rightarrow \text{Change in length, } d\ell = \frac{1}{AY} \frac{m\omega^2}{2\ell}(\ell^2 - x^2)dx$$



$$\text{Total increase in length is } \Delta\ell = \int d\ell = \int_0^\ell \frac{m\omega^2}{2AY\ell}(\ell^2 - x^2)dx = \frac{m\omega^2\ell^2}{3AY}$$

18.(1) If the rod is rotated slightly by θ , the right spring will get compressed by $\ell\theta$ and left string will get elongated by $\ell\theta$.

Restoring torque about hinge A is:

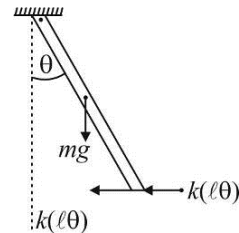
$$\tau = - \left[mg \frac{\ell}{2} \sin \theta + 2k\ell\theta(\ell \cos \theta) \right]$$

$$\because \theta \text{ is very small} \quad \Rightarrow \sin \theta = \theta, \cos \theta = 1$$

$$\therefore \tau = - \left[mg \frac{\ell}{2} + 2k\ell^2 \right] \theta$$

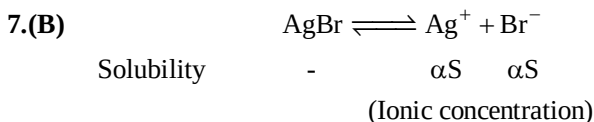
$$\Rightarrow \frac{m\ell^2}{3} \alpha = - \left[mg \frac{\ell}{2} + 2k\ell^2 \right] \theta \Rightarrow \alpha = - \left(\frac{3g}{2\ell} + \frac{6k}{m} \right) \theta = - \left[\frac{30}{2} + \frac{6(12.5)}{3} \right] \theta$$

$$\Rightarrow \alpha = -(40)\theta \quad \Rightarrow \omega^2 = 40 \quad \Rightarrow T = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{40}} = \frac{2\sqrt{10}}{2\sqrt{10}} = 1 \text{ s}$$



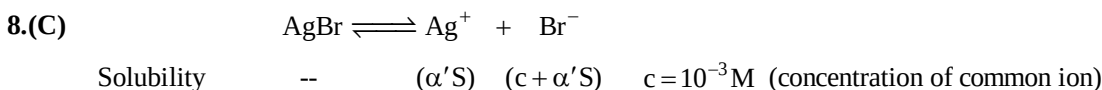
- Solutions | JEE Advanced Booster Test - 4

- 6.(ACD) (A) Reaction involve oxidation of SnO_2 into SnO_2 by the oxidant H_2O_2
 (B) Reaction involve oxidation of H_2O_2 into O_2 by Cl^{+1} (of HOCl)
 (C) Reaction involve oxidation of PbS into PbSO_4 by the oxidant H_2O_2
 (D) Reaction involve oxidation of MnO_4^{2-} into MnO_4^{-1} by the oxidant H_2O_2



$$K_{sp} = [\text{Ag}^+][\text{Br}^-] = \alpha^2 S^2 = (0.8)^2 (S)^2$$

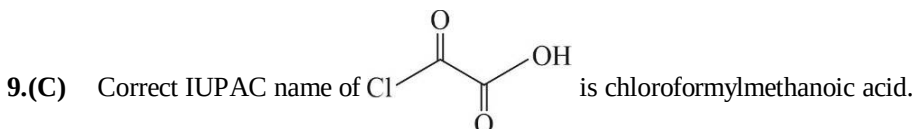
$$S = \frac{10^{-6}}{0.8} = \frac{5}{4} \times 10^{-6} = 1.25 \times 10^{-6} \text{ M}$$



$$K_{sp} = [\text{Ag}^+][\text{Br}^-] \quad (\text{neglecting } \alpha' S \text{ as compared to } c)$$

$$10^{-12} = (\alpha' S)(c) \quad (\alpha' = 0.1)$$

$$10^{-12} = 10^{-1}(S)(10^{-3}) \Rightarrow S = 10^{-8} \text{ M} \quad \therefore y = 8$$



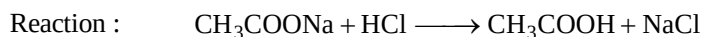
10.(C) 2, 2, 4-Trimethylpentane is the correct IUPAC name of the given compound.

11.(C) (A) 1L solution having 0.1 M CH_3COOH and 0.1 M CH_3COONa

$$\text{pH} = \text{pK}_a + \log_{10} \frac{[\text{CH}_3\text{COO}^-]}{[\text{CH}_3\text{COOH}]} = \text{pK}_a = 5$$

It will be a buffer thus pH will not change on dilution

(B) 1L solution having 0.1 M CH_3COONa and 0.1 M HCl



$$[\text{CH}_3\text{COOH}] = 0.1 \text{ M}$$

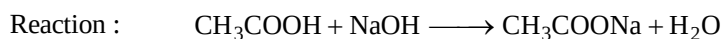
$$\text{pH} = \frac{1}{2} [\text{pK}_a - \log C] = \frac{1}{2} [5 - (-1)] = 3.0 \quad (\text{pH will change on dilution})$$

(C) 1L solution having 0.1 M HCl and 0.1 M CH_3COOH

Here due to common ion effect $[\text{H}^+]_{\text{Total}} \approx [\text{H}^+]_{\text{HCl}}$

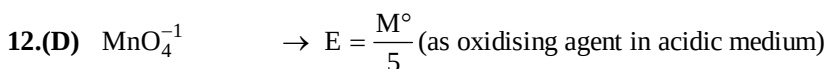
$\therefore$ pH of the solution ~ 1 (pH will change on dilution)

(D) 1L solution having 0.1 M CH_3COOH and 0.1 NaOH



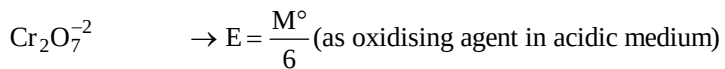
$$[\text{CH}_3\text{COONa}] = 0.1 \text{ M}$$

$$\text{pH} = 7 + \frac{1}{2} [5 + (-1)] = 7 + \frac{1}{2} [4] = 9 \quad (\text{pH of the solution will change on dilution})$$



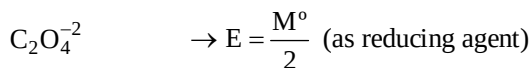
$\rightarrow$ Mn atom in its highest oxidation state (+7)

$\rightarrow$ No peroxide bond



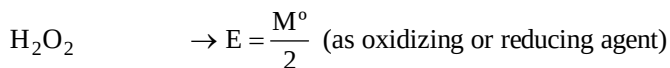
$\rightarrow$ Cr atom in its highest oxidation state (+6)

$\rightarrow$ No peroxide bond



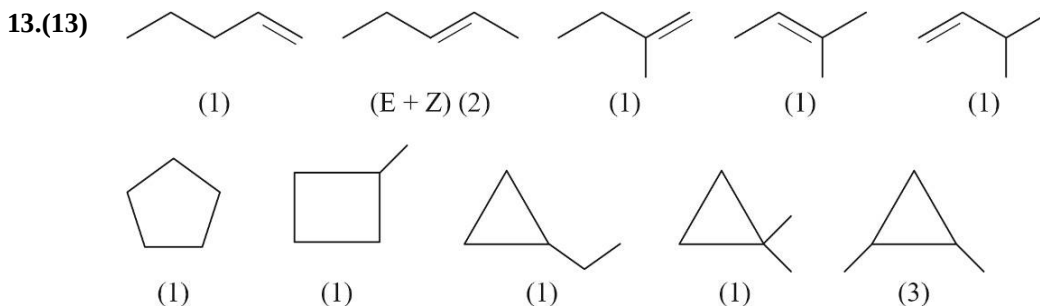
$\rightarrow$ Carbon atom in (+)3 state [Not in highest state]

$\rightarrow$ No peroxide bond



$\rightarrow$ Hydrogen atom in +1 state [Highest state]

$\rightarrow$ 1 peroxide bond



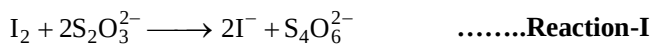
14.(9) pH of the solution at half neutralisation point = pK_a of the weak acid (HX)

$\therefore$ pK_a of the HX = 4

pH of NaX can be calculated by using

$$\text{pH} = 7 + \frac{1}{2}[\text{pK}_a + \log C] = 7 + \frac{1}{2}[4 + \log 1] = 7 + 2 = 9$$

15.(112) Reactions involved



n-factor of I^- = 6 (Reaction-II)

n-factor of O_3 = 6 (Reaction-II)

In reaction-I

Quantity of $\text{Na}_2\text{S}_2\text{O}_3$ used = 5 mmols

Quantity of I^- formed = 5 mmols

In reaction-II

Quantity of I^- used = 5 mmols

Quantity of O_3 used = 5 mmols

$V_{\text{O}_3} = 5(22.4)\text{mL} = 112\text{mL}$

16.(1) In the anion $\text{Cr}(\text{O}_2)_4^{3-}$, Cr atom is in +5 state. So n-factor of $\text{K}_2\text{Cr}_2\text{O}_7$ in reaction is 2. As also here n-factor of H_2O_2 is 2 thus 1 mole of $\text{K}_2\text{Cr}_2\text{O}_7$ can be reduced by 1 mole of H_2O_2 .

17.(5.50) pH of first equivalent point $\frac{\text{pK}_{a_1} + \text{pK}_{a_2}}{2} = 4.5$ [x]

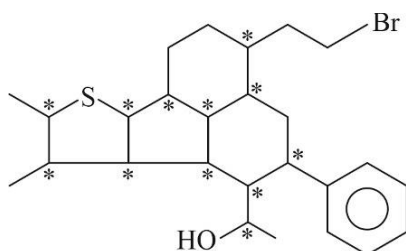
$$\text{pH of end point} = 7 + \frac{1}{2}[\text{pK}_{a_2} + \log[\text{HPO}_3^{2-}]]$$

$$\text{Here } [\text{HPO}_3^{2-}]_{\text{salt}} = \frac{100 \times 1}{300} = \frac{1}{3} \text{ M}$$

$$\therefore y = 7 + \frac{1}{2}[6.5 - 0.5] = 7 + \frac{1}{2}[6] = 10$$

$$\therefore (y - x) = 10 - 4.5 = 5.50$$

18.(12)



12 chiral atoms in the given organic compound.

[MATHEMATICS]

1.(BD) Let $A_1, A_2, \dots, A_8$ be inserted A.M.s
 $\Rightarrow 12, A_1, A_2, \dots, A_8, 75$ are in A.P with C.d = d
 $\Rightarrow 75 = 12 + (10-1)d$
 $\Rightarrow d = \frac{63}{9} = 7$
 $\therefore$

$A_1 = 19$	$A_2 = 26$	$A_3 = 33$
$A_4 = 40$	$A_5 = 47$	$A_6 = 54$
$A_7 = 61$	$A_8 = 68$	

2.(AD) $y = \frac{3\sin x - 4\sin^3 x}{\sin x} = 3 - 4\sin^2 x$
 $0 < \sin^2 x \leq 1 \quad (\sin x \neq 0)$
 $\Rightarrow 3 > y \geq -1$

3.(BC) Option (B)

$$\frac{\left[\sin\left(\frac{3 \times 2\pi}{7 \times 2}\right) \right] \cos\left[\frac{\frac{2\pi}{7} + \frac{6\pi}{7}}{2}\right]}{\sin\left(\frac{2\pi}{7 \times 2}\right)} = \frac{\sin\frac{4\pi}{7} \cdot \cos\frac{4\pi}{7}}{\sin\frac{\pi}{7}} = \frac{\sin\frac{4\pi}{7} \cdot \cos\frac{4\pi}{7}}{\sin\frac{\pi}{7}} = \frac{1}{2} \frac{\sin\frac{8\pi}{7}}{\sin\frac{\pi}{7}} = \frac{1}{2} \frac{\sin\left(\pi + \frac{\pi}{7}\right)}{\sin\frac{\pi}{7}} = -\frac{1}{2}$$

Option (C)

$$\cos\frac{2\pi}{7} \cdot \cos\frac{4\pi}{7} \cdot \cos\frac{8\pi}{7} = \frac{1}{2^3} \therefore \frac{\sin\frac{16\pi}{7}}{\sin\frac{2\pi}{7}} = \frac{1}{8}$$

4.(AC) $a = 3, b = 2$

$$\text{L.R.} = \frac{2b^2}{a} = 2 \times \frac{4}{3} = \frac{8}{3}$$

$$e^2 = 1 - \frac{b^2}{a^2} = 1 - \frac{4}{9} = \frac{5}{9}$$

$$e = \frac{\sqrt{5}}{3} \quad ae = \sqrt{5}$$

$$\text{Dist. between foci} = 2ae = 2\sqrt{5} \quad \text{Dist. Between directrices} = \frac{2a}{e} = \frac{2 \times 3 \times 3}{\sqrt{5}} = \frac{18}{\sqrt{5}}$$

5.(ABD) Let the tangent to parabola be $y = mx + \frac{2}{m}$

$$\text{Apply condition of tangency w.r.t circle, } \frac{\left|\frac{2}{m}\right|}{\sqrt{m^2 + 1}} = \sqrt{2} \Rightarrow \frac{2}{m^2(m^2 + 1)} = 1$$

$$\Rightarrow m^4 + m^2 - 2 = 0 \Rightarrow (m^2 + 2)(m^2 - 1) = 0 \Rightarrow m = \pm 1$$

Consider $m = 1 \quad y = x + 2$

Solve with $x^2 + y^2 = 2$

$$\Rightarrow x^2 + x^2 + 4 + 4x = 2 \Rightarrow 2x^2 + 4x + 2 = 0 \Rightarrow x^2 + 2x + 1 = 0 \Rightarrow x = -1, y = 1$$

$\therefore P \equiv (-1, 1)$ Similarly, we can find other points

6.(AC) $\frac{a+b+c}{3} = b+2$ & $b^2 = ac$

$\Rightarrow a+b+c = 3b+6$

$\Rightarrow a+c = 2b+6$ (1)

Let $b = ar$ $c = ar^2$

(1) becomes

$a+ar^2 = 2ar+6 \Rightarrow a[r^2 - 2a+1] = 6 \Rightarrow a(r-1)^2 = 6$

a is integer & r is integer

$\Rightarrow a = 6$ & $(r-1)^2 = 1 \Rightarrow r = 2$ [$r \neq 0$]

7.(D)-8.(B)

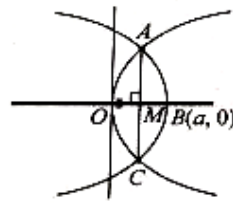
Solving equations of parabolas we get points of intersection as $A\left(\frac{2a}{3}, \sqrt{\frac{8a}{3}}\right)$ and $C\left(\frac{2a}{3}, -\sqrt{\frac{8a}{3}}\right)$

But points O, A, B, C are concyclic.

So, by the figure, we can say $OM \times MB = AM \times MC$

$\Rightarrow \frac{2a}{3} \times \left(a - \frac{2a}{3}\right) = \sqrt{\frac{8a}{3}} \times \sqrt{\frac{8a}{3}}$

$\frac{2a}{3} \times \frac{a}{3} = \frac{8a}{3} \Rightarrow a = 12$



So, length of common chord $= 2\sqrt{\frac{8a}{3}} = 8\sqrt{2}$

And Area of quadrilateral OABC $= 2 \times ar(\Delta OAB) = 2 \times \frac{1}{2} \times a \times \sqrt{\frac{8a}{3}} = 48\sqrt{2}$

9.(C)

Case-I All odd $2 \times 3 \times 4 = 24$

Case-II x_1, x_2 even $1 \times 2 \times 4 = 8$

x_3 odd

Case-III x_1, x_3 even $1 \times 3 \times 3 = 9$

x_2 odd

Case-IV x_2, x_3 even $2 \times 2 \times 3 = 12$

x_1 odd

53

10.(D) $x_1 + x_3 = 2x_2 \Rightarrow x_1 + x_3$ is even

Case - I x_1, x_3 even $1 \times 3 = 3$

Case - II x_1, x_3 odd $2 \times 4 = 8$

11

11.(A) (A) Tangent at the each of focal chord meet at 90° and meet at directrix

(B) The angle between SP and SQ is 90°

(C) Condition for focal chord $t_1 t_2 = -1$

(D) Condition for normal chord

$$12.(C) \quad (A) \quad T_n = n(n+1)(n+1)(n+2) \Rightarrow S_n = \frac{n(n+1)(n+2)(n+3)(n+4)}{5} + C$$

At $n=1$

$$1 \cdot 2 \cdot 3 \cdot 4 = \frac{1 \times 2 \times 3 \times 4 \times 5}{5} + C$$

$$\Rightarrow C = 0 \Rightarrow S_{11} = \frac{11 \times 12 \times 13 \times 14 \times 15}{5} = 72072$$

$$(B) \quad T_n = \frac{1^3 + 2^3 + \dots + n^3}{1 + 3 + 5 + \dots + (2n-1)} = \frac{\left(\frac{n(n+1)}{2}\right)^2}{n^2} = \frac{1}{4}(n+1)^2$$

$$T_n = \frac{1}{4}(n+1)^2$$

$$S_{16} = \frac{1}{4}[2^2 + 3^2 + 4^2 + \dots + 17^2]$$

$$= \frac{1}{4}[(1^2 + 2^2 + \dots + 17^2) - 1] = \frac{1}{4}\left[\frac{17 \times 18 \times 35}{6} - 1\right] = \frac{1}{4}[1785 - 1] = \frac{1784}{4} = 446$$

$$(C) \quad T_n = n(n+2) = n^2 + 2n$$

$$S_n = \Sigma n^2 + \Sigma 2n$$

$$S_n = \frac{n(n+1)(2n+1)}{6} + n(n+1)$$

$$S_{30} = \frac{30 \times 31 \times 61}{6} + 30 \times 31 = \frac{5 \times 31 \times 61}{6} + 30 \times 31 = 10385$$

$$(D) \quad T_n = \frac{n(n+1)}{2} = \frac{1}{2}(n^2 + n)$$

$$S_n = \frac{1}{2}[\Sigma n^2 + \Sigma n]$$

$$S_n = \frac{1}{2}\left[\frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2}\right] = \frac{n(n+1)(n+2)}{6} \Rightarrow S_{60} = \frac{60 \times 61 \times 62}{6} = 37820$$

$$13.(3) \quad \text{Coefficient of } x^n \text{ in } (1+x^2)^4(1+x^3)^7(1+x^4)^{12}$$

$$= \text{Coefficient of } x^n \text{ in } ({}^4C_0 + {}^4C_1x^2 + {}^4C_2x^4 + {}^4C_3x^6 + {}^4C_4x^8) \times$$

$$(1 + {}^7C_1x^3 + {}^7C_2x^6 + {}^7C_3x^9 + \dots) (1 + {}^{12}C_1x^4 + {}^{12}C_2x^8 + \dots)$$

$$= {}^7C_1 \cdot {}^{12}C_2 + {}^4C_1 \cdot {}^7C_3 + {}^4C_2 \cdot {}^7C_1 + {}^4C_2 \cdot {}^{12}C_1 + {}^4C_4 \cdot {}^7C_1 = 1113$$

$$14.(7) \quad {}^nC_0 - {}^nC_1 + {}^nC_2 - \dots + (-1)^{m-1} {}^nC_{m-1}$$

$$= \text{constant term in } \left[{}^nC_0 - {}^nC_1x + {}^nC_2x^2 - \dots + (-1)^n {}^nC_nx^n \right] \left[1 + \frac{1}{x} + \frac{1}{x^2} + \dots + \frac{1}{x^{m-1}} \right]$$

$$= \text{constant term in } (1-x)^n \left(\frac{1 - \frac{1}{x^m}}{1 - \frac{1}{x}} \right) = \text{constant term in } (1-x)^n \frac{(1-x^m)}{x^{m-1}(1-x)}$$

$$= \text{coefficient of } x^{m-1} \text{ in } (1-x)^{n-1}(1-x^m)$$

$$x^{m-1} \text{ in } (1-x^{n-1}) = (-1)^{m-1} ({}^{n-1}C_{m-1}) \Rightarrow f(7, 8) = {}^7C_6 = 7$$

15.(3) Number of blue segments = n

Number of red segments = ${}^nC_2 - n$

As red & blue segments are equal

$${}^nC_2 - n = n \Rightarrow {}^nC_2 = 2n$$

$$\Rightarrow \frac{n(n-1)}{2} = 2n \Rightarrow n-1=4 \Rightarrow n=5$$

16.(4) $\cos^4 x + \sin^4 x = 1 - 2\sin^2 x \cdot \cos^2 x$

$$\cos^6 x + \sin^6 x = 1 - 3\sin^2 x \cdot \cos^2 x$$

Equation becomes

$$\frac{5}{4}\cos^2 2x + 2 - 5\sin^2 x = 2$$

$$\Rightarrow \frac{5}{4}[\cos^2 2x - \sin^2 2x] = 0 \Rightarrow \cos 4x = 0$$

$$\text{Period} = \frac{\pi}{2}$$

$\therefore$ 4 solutions in $x \in [0, \pi]$

$$17.(7) T_r = \frac{1+2+\dots+r}{\sqrt{1+3+5+(2r-1)}} = \frac{r(r+1)}{2 \times \sqrt{r^2}}$$

$$S_9 = \sum_{r=1}^9 \left(\frac{r+1}{2} \right)$$

$$18.(9) \frac{a^{-4} + a^{-3} + a^{-2} + a^{-2} + a^{-2} + a^{-2} + 1 + a^7 + a^8}{9} \geq 1$$

Equality when $a = 1$